

How Frequency Band Importance Affects Neural Network Predictions and Human Perception for Speech Quality Assessment

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Abstract

Audio quality assessment models are often used for evaluating speech enhancement and noise suppression approaches. However, recent work has shown that network predictions do not always correlate well with human predictions. Explainability could help developers and users of these tools identify potential error and data bias, which would expedite development. In this work, we use Shapley additive explanations, partial dependence plots, and perturbation analysis to determine the influential frequency bands for three quality assessment models. We then conduct a listening study to determine if human perception is sensitive to the same factors as these networks. Our results show that both networks and humans are more sensitive to detrimental factors than beneficial ones. Further, the networks focus on a small portion of the input which corresponds to low frequency bands. Our work represents a step towards unifying explainability for both humans and networks.

Index Terms: speech assessment, quality assessment, explainability, human perception

1. Introduction

Perceptual speech quality refers to how well a signal is perceived and can be highly subjective. A common metric is the mean opinion score (MOS) of a signal, determined by the average of human-rated quality scores (1 - very bad, 5 - very good). Deep learning approaches are commonly used for automated quality assessment and leverage a variety of architectures including convolutional networks [1, 2, 3, 4] to those with self-attention [5] or (bidirectional) long short term memory [3, 6, 7]. Recent works have also utilized semi- [8] and self-supervised learning [1, 9].

While DNN approaches achieve competitive accuracies, the networks' predictions do not always correlate well with human assessments [7, 10, 11], especially on unseen data [12, 13, 14]. As a result, investigation into how these networks make their assessments is needed. Understanding the cause of this discrepancy informs developers of the root cause of error instead of blindly trying to improve accuracy. Other work in speech explainability has been limited to either one model or one explanation method which severely limits the generalizability of these explanations [15, 16, 17, 18, 19, 20]. Additionally, few works incorporate human-based studies. We aim to further our understanding of DNN decision-making for speech quality assessment by looking at several explanation approaches for multiple networks and incorporating human perception through the use of listening studies to determine the most influential frequency bands for speech quality prediction.

Explanations can be local or global. A local explanation elucidates model prediction for a single input while global explanations describe model behavior over a set of inputs [16, 19, 21, 22, 23, 24, 25]. Local explanations are especially helpful in post-hoc scenarios. For example, given a model that takes patient medical symptoms as input and predicts a diagnosis, a doctor may use local explanations to see what led the model to its diagnosis *in this particular case*. Thus, local explanations cannot illuminate general network trends. On the other hand, global explanations assess an entire system and can evaluate, for example, the *overall* fairness of a model with respect to certain sensitive attributes. Compared to local, global explanations provide a fuller picture of network behavior and can be used for determining systemic issues (e.g., bias).

Global explanations are categorized as model-driven or data-driven. Data-driven methods describe how specific inputs impact model predictions, whereas model-driven methods focus on a model's internal mechanisms (e.g., weights, activations) [21, 26, 27, 28, 29, 30]. Gradient-based approaches [16, 22, 23, 24, 25, 31] use partial derivatives to assign relevance to neurons. Layer-wise Relevance Propagation (LRP) [16] and Spectral Relevance Analysis (SpRAy) [31] attribute relevance scores to input features during back-propagation which are then visualized as heatmaps. LRP's local explanations can be aggregated into pseudo-global ones, while SpRAy's qualitative explanations require human interpretation [16, 31, 32].

Perturbation-based methods are a common approach to data-driven explanations [26, 27, 28, 33]. Activation maximization finds patterns in the input data that maximize the activation function for a given neuron [28]. Other global explanations aggregate a set of local explanations. SP-LIME learns an interpretable model which is approximately correct in the neighborhood of given data point and then aggregates over a set of inputs [21]. Two limitations to SP-LIME are that an optimal solution is not guaranteed and it requires a locally-approximate model to be trained for each input which is computationally expensive. Global attribution analysis uses several local explanations to determine important input features and then normalizes and clusters these attributions to achieve a global explanation [33]. Similar to SP-LIME, the aggregation of local explanations is computationally expensive as these approaches have to be repeated for many inputs and are ultimately sensitive to sample selection. Shapley values [34] use game theory to assign a contribution of each "player" (i.e., feature, input signal, etc.) to the outcome and have been used for evaluating audio prediction networks [35, 36]. However, the focus is typically on data valuation and using Shapley to curate a dataset without drawing conclusions on the network's behavior with respect to its input features.

While there has been significant work in creating expla-

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nation frameworks, studies proposing new approaches tend to evaluate only their work on several networks, limiting the conclusions to their approach [16, 17, 18, 21]. Conversely, works that aim to evaluate explanations tend to focus on several approaches for one network, restricting their conclusions to that network [15, 19, 20]. In this paper, we use SHAP [27], partial dependence plots (PDP) [26], and perturbation analysis to study the influential frequency bands for MOSNet [3], DNSMOS [2], and SCOREQ [12]. Common architectures for audio quality assessment rely on convolution or attention [37]. Attention-based networks are not as easily interpretable [38, 39] due to their increased complexity, making convolutional networks a good stepping stone. DNSMOS and MOSNet are both convolutional networks that predict the mean opinion score of audio signals. They are well-known within the quality assessment field and were state-of-the-art at publication, providing a good baseline. SCOREQ is a recent state-of-the-art which utilizes a more complex architecture. Both SHAP and PDP have the advantage of being easily interpretable and quantitative [26, 27, 32].

We present one of the first studies to investigate which frequency bands influence speech quality across multiple networks and approaches, while also comparing these results to human perception. By investigating the determining factors for both networks and humans, we help illuminate the cause of discrepancy between human and network predictions. If networks are focusing on different frequency bands than humans, then the lack of correlation may be due to a different decision method. Otherwise, the difference could simply be due to engineering parameters. This work guides future AI development by using human behavior to identify whether neural networks are following human perception trends during decision-making.

2. Neural Network Explainability

2.1. Explanation Methods

Let f be the model's prediction function that we are trying to explain and suppose we have a set of inputs, $\{\mathbf{x}^i\}_{i=1}^N$ where each $\mathbf{x}^i = \{x_m^i\}_{m \in M}$ is comprised of $|M|$ features. Consider an arbitrary feature m and the pair $(\{x_m^i\}_{i=1}^N, \{x_{M \setminus m}^i\}_{i=1}^N)$ where $x_m^i \cup x_{M \setminus m}^i = \mathbf{x}^i$. For brevity, we abbreviate $\{x_m^i\}_{i=1}^N$ and $\{x_{M \setminus m}^i\}_{i=1}^N$ as $\mathbf{x}_m$ and $\mathbf{x}_{M \setminus m}$, respectively.

2.1.1. SHAP

SHAP (SHapley Additive exPlanation) [27] is a local explanation approach that measures the effects of a specific feature, m , on model prediction for a single data instance, $\mathbf{x}$. The SHAP value for feature m , ϕ_m , is calculated by retraining the model on all possible subsets of features excluding m , denoted $S \subset M \setminus m$ where M is the set of all features. The Shapley value for feature m is then the weighted average of the difference between the model's output including the feature, $f_{S \cup \{m\}}(\mathbf{x})$, and excluding the feature, $f_S(\mathbf{x})$:

$$\phi_m = \sum_{S \subset M \setminus m} \frac{|S|!(|M|-|S|-1)!}{|M|!} (f_{S \cup \{m\}}(\mathbf{x}_{S \cup \{m\}}) - f_S(\mathbf{x}_S)) \quad (1)$$

To be "additive", the sum of SHAP values for all features must approximate the output $f(\mathbf{x})$ of the model being explained. SHAP values are an implementation of Shapley values such that SHAP (1) is locally accurate: the output of the explanation model must match the output of the original model; (2) has missingness: if a feature is not present in the input, it has

a SHAP value of 0; and (3) is consistent: if a contributing feature's value in an input stays the same or increases, the SHAP value for that feature should not decrease. The equations for each of these properties can be found in [27].

2.1.2. Partial Dependence Plots

Partial dependence plots (PDPs) summarize how the model's prediction, f , changes in response to different input values of a specified feature, m . We describe the partial dependence of $f(\mathbf{x})$ on m , denoted as $\hat{f}_m(\mathbf{x}_m)$, by the expectation of $f(\mathbf{x})$ over the inputs excluding feature m .

$$\hat{f}_m(\mathbf{x}_m) = \mathbb{E}_{\mathbf{x}_{M \setminus m}} [f(\mathbf{x})] \quad (2)$$

$$= \int f(\mathbf{x}_m, \mathbf{x}_{M \setminus m}) p_{\mathbf{x}_{M \setminus m}}(\mathbf{x}_{M \setminus m}) d\mathbf{x}_{M \setminus m} \quad (3)$$

where $p_{\mathbf{x}_{M \setminus m}}(\mathbf{x}_{M \setminus m})$ is the marginal probability density of $\mathbf{x}_{M \setminus m}$, i.e., $p_{\mathbf{x}_{M \setminus m}}(\mathbf{x}_{M \setminus m}) = \int p(\mathbf{x}) d\mathbf{x}_m$, where $p(\mathbf{x})$ is the joint density for all input features. $p_{\mathbf{x}_{M \setminus m}}(\mathbf{x}_{M \setminus m})$ can be estimated by averaging $f(\mathbf{x}_m, \mathbf{x}_{M \setminus m})$ over each data point in $\mathbf{x}_{M \setminus m}$. This allows us to rewrite Equation (3) as

$$\hat{f}_m(\mathbf{x}_m) = \frac{1}{N} \sum_{i=1}^N f(\mathbf{x}_m, \mathbf{x}_{M \setminus m}^i) \quad (4)$$

The PDP curve for feature m results from evaluating $f(\mathbf{x}_m, \mathbf{x}_{M \setminus m})$ and averaging out each feature in $\mathbf{x}_{M \setminus m}$.

2.1.3. Perturbation

In perturbation (or sensitivity) analysis, we measure the effect of an input variable by making a minimal change and observing the change in the resulting output. In our experiments, we investigate how modifying the average magnitude of a frequency band impacts model prediction of speech quality. For each input signal, we select a single frequency band and scale the power at each time frame by a constant factor (0.25, 0.5, 10, 50, 100, or 1000). The perturbed signals were then converted back to the time domain using the same phase as the original signal. Only one frequency band at a time is altered, and we discuss the frequency bands chosen for modification in section 2.4.

2.2. Models

MOSNet-CNN [3] (hereafter referred to as MOSNet) is a 17-layer convolutional regression network that predicts the mean opinion score of a signal. Using the original implementation¹, we retrained MOSNet to take as input a magnitude spectrogram with 161 frequency bands (as opposed to the original 257). The output of the model is the predicted utterance-level MOS, although frame-wise scores can be taken from the penultimate layer. MOSNet was retrained to accept 161 frequency bands as input to promote comparison with other networks in this work as each has features corresponding to 50 Hz-wide frequency bands. Retraining MOSNet to accept an input spectrogram with 161 frequency bands instead of 257 did not lead to substantial performance changes. MOSNet was trained on VCC-2018 [40] and the original authors' network weights were used.

DNSMOS [2] is a 14-layer convolutional regression model that predicts speech quality, background noise quality, and overall audio quality. For this work, we are only interested in

¹<https://github.com/lochenchou/MOSNet>

Table 1: Mean squared error (MSE), linear correlation coefficient (LCC), and Spearman rank correlation coefficient (SRCC) performance baselines for DNSMOS, MOSNet, SCOREQ on the IUCOSINE test set.

Model	MSE ($\downarrow$)	LCC ($\uparrow$)	SRCC ($\uparrow$)
DNSMOS	0.3890	0.2317	0.1981
MOSNet	0.5248	0.1877	0.1981
SCOREQ	0.4624	0.6283	0.6249

the overall audio quality (MOS) prediction. We use the original implementation and network weights² of DNSMOS which was trained on the DNS-Challenge dataset [41] and takes a log power spectrogram with 161 frequency bands as input.

SCOREQ [12] is a variant of the wav2vec 2.0 architecture [42] which utilizes a triplet loss and contrastive training to finetune the wav2vec 2.0 transformer. Wav2vec 2.0 is comprised of a multi-layer convolutional feature encoder followed by a transformer and quantization module. SCOREQ follows a two-step approach in which the encoder and a projection head are first trained using the SCOREQ triplet loss. In the next step, wav2vec 2.0’s encoder and transformer are frozen while an L2 loss trains a linear MOS prediction head. SCOREQ was trained on the NISQA TRAIN SIM set [5] and the original authors’ implementation and network weights were used³.

2.3. Dataset

All models in this work were trained on the data used in the original implementations. To prevent any network from having the unfair advantage in our analyses of being tested on a dataset seen during training, we use a speech quality dataset that does not overlap with any dataset the models were trained on. Dong and Williamson [43] performed a human-subject listening study to collect MOS ratings for audio clips from the CONversational Speech In Noisy Environments (COSINE) corpus [44]. COSINE is a real-world dataset generated by securing microphones in four places (mouth, shoulder, throat, chest) on human subjects and recording audio as the subjects go about their day. We refer to this dataset as IUCOSINE, and it contains 14,400 training samples and 3,600 test samples. The training samples were used for instantiating and training the prediction models and the test set was used for evaluation. To form the evaluation set, we randomly selected 100 signals from the IUCOSINE test set such that the distribution of true MOS for our evaluation set matches that of the IUCOSINE test set.

Table 1 shows the baseline performance of each model on the IUCOSINE dataset. All experiments were run on a 48-core Dual Intel Xeon 8268s or NVIDIA Volta V100 GPU [45]. One of the motivations for this work is the lack of correlation between human and network predictions [7, 10, 11], which is exhibited in the lower-than-ideal LCC and SRCC values.

2.4. Selected Frequency Bands

Though the models’ input features span 161 frequency bands, each with a width of 50 Hz, for brevity we focus our discussion on just nine. We concentrate on these bands as they include key perceptual features (such as fundamental pitch) or

were chosen empirically from our analyses. All models were evaluated on six common frequency bands: 3–5 (150–300 Hz), 31 (1550–1600 Hz), and 40–41 (2000–2500 Hz). Bands 3–5 encompass the modal range of young men and women ($85\text{--}155\text{ Hz} \pm 2.11\text{ semi-tones}$ and $165\text{--}255\text{ Hz} \pm 1.7\text{ semitones}$, respectively) [46]. Band 31 was identified by Apoux and Healy [47] as having high relative band importance for consonants. Bands 40–41 were similarly identified as highly important for vowels.

The remaining three frequency bands of interest were chosen separately for each model based on the bands that reported the highest average magnitude of SHAP values (i.e., the frequency bands which had the largest average impact on model prediction), as discussed in section 4.1 and Fig. 1. The empirically-chosen frequency bands are (14, 15, 26) for DNSMOS, (6, 12, 17) for MOSNet, and (2, 20, 21) for SCOREQ.

3. Human Perception

Given that speech quality assessment aims to match human perception, it follows that humans and networks should focus on the same factors. Therefore, we conduct a listening study to determine if network behavior aligns with human perception. For this IRB-approved study, 200 participants were asked to rate the quality of signals that are organized into (original, perturbed) pairs, give their preference (if any) within each pair, and indicate which factors affected their quality ratings. Preference tests, as opposed to a single-utterance test, are widely used to facilitate comparisons between multiple situations (for example, when comparing the quality of two signals with different noise levels) and have been found to be more reliable than single-score tests [48, 49, 50].

We perturb eight randomly-selected signals from our evaluation set, discussed in section 2.3. Of the eight files selected, there is one signal each with true MOS 1 and 4, and three signals each with true MOS 2 and 3. Four of the eight signals have female speakers and four have male speakers. The eight selected signals were perturbed by scaling one of the selected frequency bands (section 2.4) by 0.5, 10, or 1000. The scaling factors were chosen empirically from the results in Figs. 3a, 3c, and 3e as 0.5, 10, and 1000 predominantly define the shape of the curves showing the effect of perturbation on neural network behavior (section 4.3). The listening study presents one pair of signals in each question. Each pair contains, in random order, the original, unmodified audio signal and the same audio signal with one frequency band perturbed to a specific scale. We refer to the four pairs associated with a specific frequency band as a “block.” The pair (source signal, source signal) is also included to serve as an anchor question. Thus, for each source signal, there are 60 associated pairs (four scales, 15 frequency bands). Each participant is randomly assigned four blocks to rate (16 pairs), though the pairs are presented in random order. After rating the quality of each signal in the pair on a 5-point scale, participants are asked to select which signal they prefer.

Finally, participants are presented with four factors and asked to indicate if the factor contributed negatively, positively, or did not affect their rating of their preferred signal. The four factors given are the amount of noise, loudness of speech compared to noise, audio distortion, and type of noise. Each of these factors were chosen to correspond to an acoustic property which is known to affect quality ratings [51, 52] and is recognizable within a spectrogram.

Participants were recruited via Prolific and screened to ensure all are fluent in English, have lived in the United States for at least 5 years, and have self-reported normal hearing. The sur-

²<https://github.com/microsoft/DNS-Challenge>

³<https://github.com/alessandroragano/scoreq>

vey was administered using Qualtrics. The mean survey duration was 17.9 minutes with a standard deviation of 7.5 minutes. 99 participants were female and 101 were male. The majority of participants (131) were in the range 30-49 years of age and predominantly identified as White (143).

4. Experiments

4.1. SHAP

SHAP values quantify the influence of an input feature on a model's average prediction. To determine which frequency band has the strongest affect on model prediction, we calculate the SHAP values for each of the 100 files in our evaluation set, shown in Fig. 1. Positive SHAP values for a frequency band indicate that inclusion of that frequency band increases the model's average prediction; negative SHAP values indicate that inclusion of the frequency band lowers the average model prediction. The actual SHAP value is not informative on its own but rather it's the sign and magnitude compared to other SHAP values that matter. For example, in Fig. 1a, the average SHAP value (blue bar) for frequency band 4 has a large, positive value compared to the negative value of frequency band 0 or smaller positive value of frequency band 3. Thus, including frequency bands 3 and 4 as input features increases DNSMOS's average MOS prediction while including frequency band 0 decreases it, where frequency band 4 increases the average prediction more than frequency band 3 does.

The SHAP values for MOSNet (Fig. 1b), show that the lower frequency bands are the most influential with the higher frequency bands ($>$ band 40) having very little effect on prediction. Though less severe, DNSMOS and SCOREQ show similar behaviors with the lower bands ($<$ band 40) having stronger influence. Unlike MOSNet, DNSMOS (Fig. 1a) and SCOREQ (Fig. 1c) are affected by almost all bands (i.e., having $|\text{SHAP}| > 0$). While MOSNet is largely unaffected by the upper bands, for DNSMOS, a strong majority of these upper bands have a negative effect and for SCOREQ, the majority of bands have a positive effect (though lesser than that of the lower frequency bands). In fact, DNSMOS having the clearest separation of low frequency bands ($<$ band 40) being positively influential compared to the negative influence of the upper frequency bands may explain DNSMOS's superior performance (Tab. 1) as it has been suggested by other works that lower frequency bands are important for speech perception [53].

The left plots in Fig. 2 are an alternative presentation of the SHAP data in Fig. 1. Instead of using the average SHAP value for each frequency band, we show the individual SHAP values for a single band, in this case, band 40. Each scatter point shows the SHAP value for band 40 compared to the band's average magnitude across all time frames for each signal. The shape and color of the scatter points indicate the (truncated) MOS value that the model will ultimately predict for the signal. Though we do not see a clear separation between SHAP value or even average magnitude and predicted MOS, we plot the trendline of the SHAP values as the black dotted line, which we use to compare to the PDP results in the following section.

4.2. PDP

Next, we determine if the behavior from the SHAP explanation (local) matches the behavior reported by PDP (global). The PDP curves, computed across the entire IUCOSINE dataset, are shown by the right-side figures in Fig. 2. The PDP curves show that for DNSMOS (Fig. 2a) and SCOREQ (Fig. 2c), average

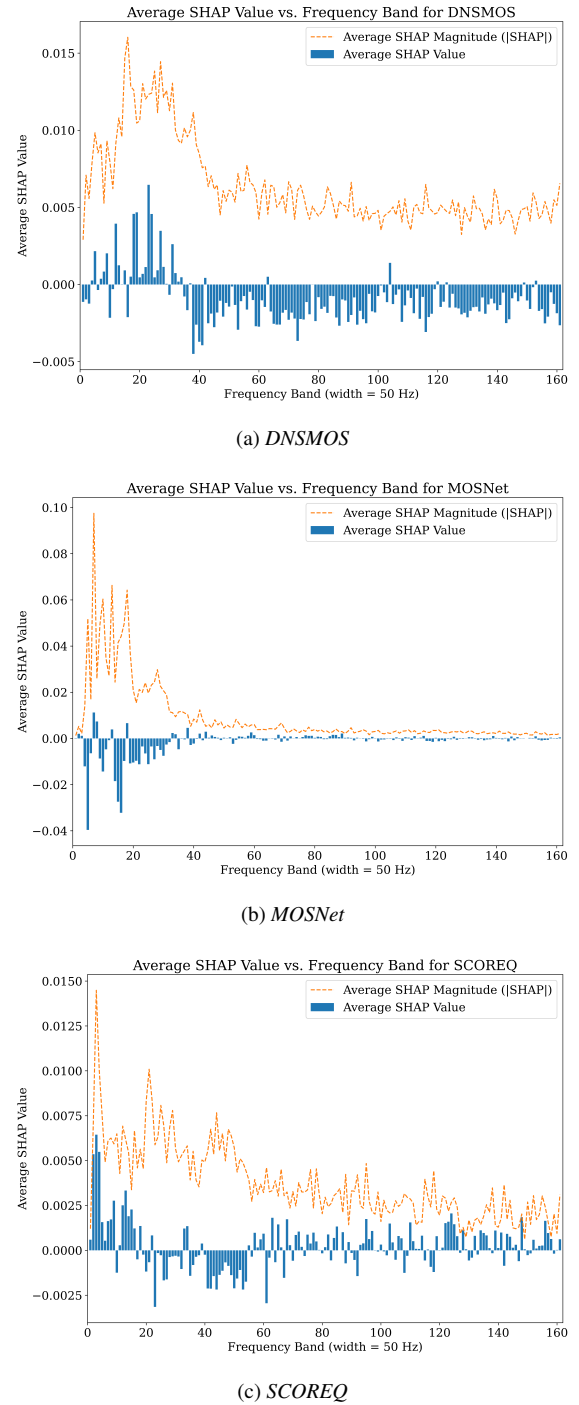
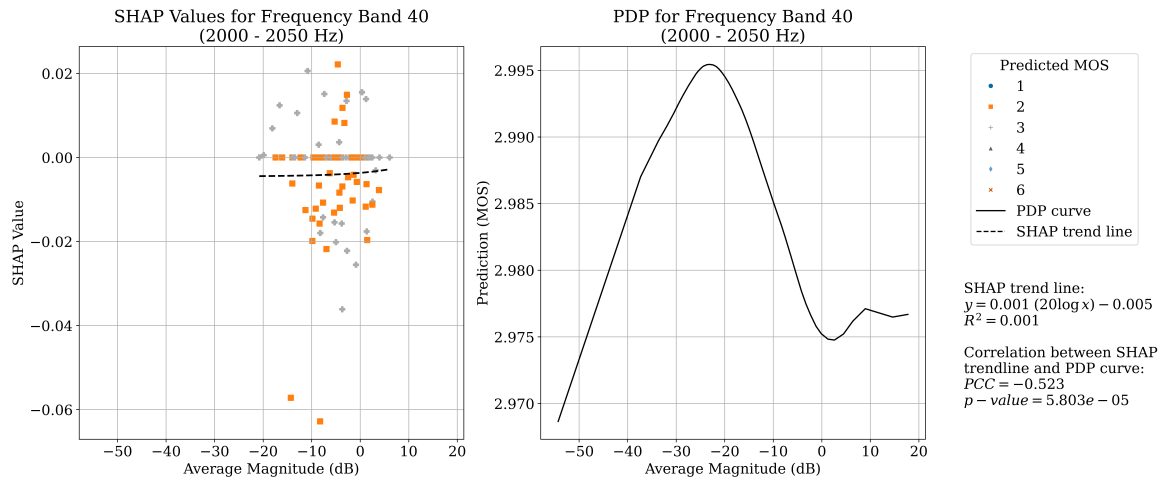
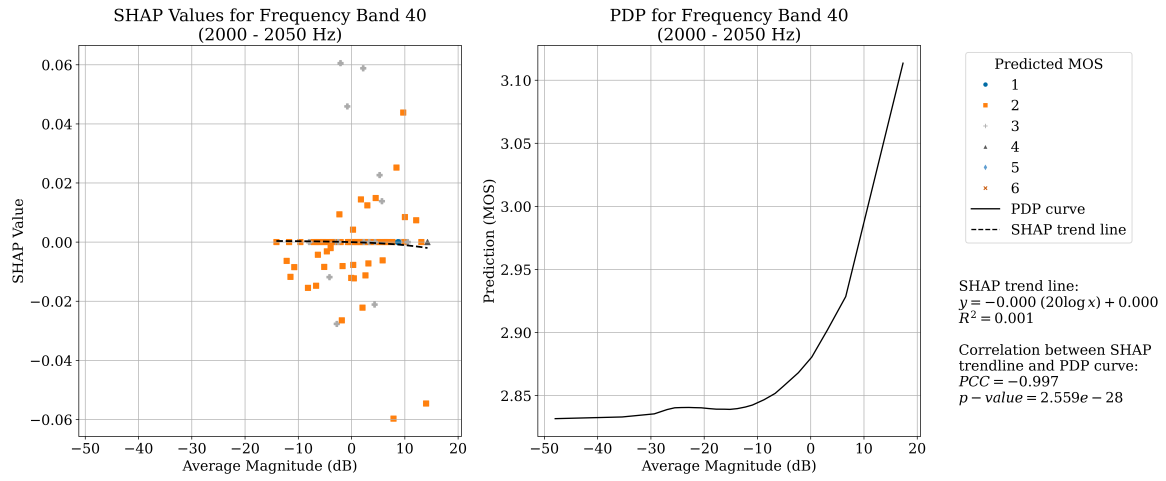


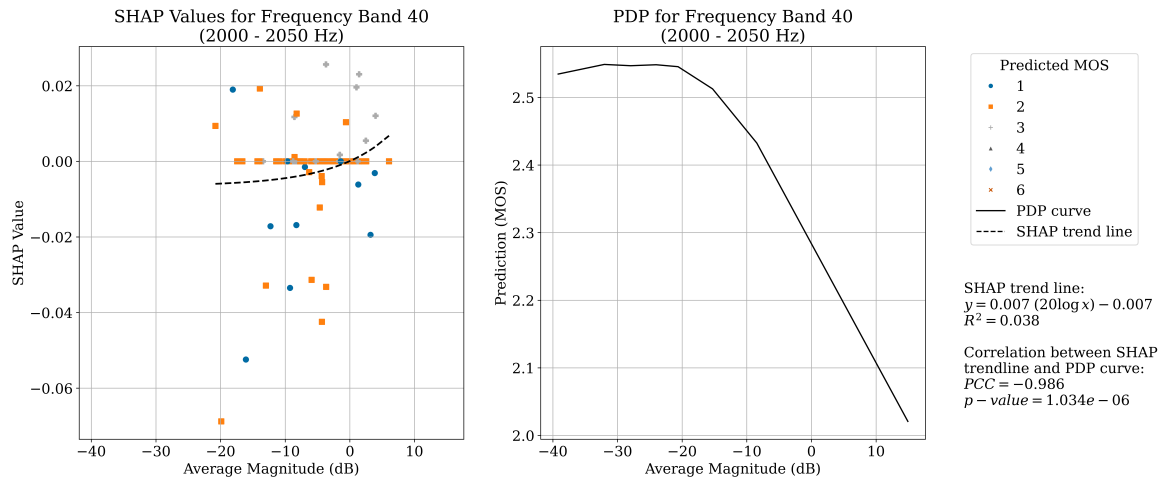
Figure 1: Average SHAP values and average magnitude of SHAP values by frequency band for DNSMOS, MOSNet, and SCOREQ across 100 randomly selected test files. The bars indicate the average SHAP value for a given frequency band: negative values indicate that the inclusion of the feature, on average, decreases model prediction while positive values indicate an increase in model prediction. The dotted line shows the average magnitude of SHAP values across each band. Larger magnitudes correspond to stronger influence.



(a) *DNSMOS*



(b) *MOSNet*



(c) *SCOREQ*

Figure 2: SHAP values (scatter points, left side of figure, colored by predicted MOS of the file) of 100 randomly-selected test files and PDP curves (right) for *DNSMOS*, *MOSNet*, and *SCOREQ*. The Pearson's correlation coefficient between the trendline of SHAP values and PDP curve is given by the text on the right.

magnitudes of approximately -20 dB or higher in frequency band 40 decrease MOS predictions while MOSNet (Fig. 1b) predicts progressively higher MOS ratings as the average magnitude of increases.

The text on the right side of the figures shows the correlation between the SHAP value trendline and the PDP curve for each model. The correlation indicates whether the SHAP and PDP explanations agree. Though the PDP curves were calculated for each of the nine frequency bands of interest, we only discuss one band for each model here for brevity. Despite seeing similar trends in other frequency bands, we chose frequency band 40 for discussion as it has the strongest (positive or negative) correlation between SHAP and PDP for both MOSNet and SCOREQ. For DNSMOS, the strongest correlation was for frequency band 15, which is also the frequency band that was reported by SHAP to have the strongest influence.

For both DNSMOS and SCOREQ, the trend of the SHAP values disagrees with that of PDP, shown by the negative correlations. However, most of the SHAP values are zero which causes the trendline to fit poorly. In Fig. 2a, most SHAP values are negative which agrees both with Figure 1a and the negative slope of the PDP curve for magnitudes in the range $[-20, 0]$. MOSNet, however, shows strong agreement between the SHAP values and PDP curve in Fig. 2b and the results in Fig. 1b.

4.3. Perturbation Analysis

The SHAP and PDP results from section 4.1 indicate that increasing the average magnitude in frequency band 40 should result in a higher MOS prediction by MOSNet. To confirm this behavior, for each of the 100 signals used in the previous experiments, we scale each frequency band of interest by $[0.25, 0.5, 10, 50, 100, 1000]$. Figures 3a, 3c, and 3e show how the predictions change with these scale factors.

DNSMOS (Fig. 3a) has the most stable behavior with its predictions staying in the 2–3 MOS range. In general, increasing frequency magnitudes resulted in lower predictions by DNSMOS. However, there is a slight increase when the upper frequencies (26, 31, 40, and 41) are scaled by a factor of 10. In each case, the perturbation results match those from SHAP and PDP, indicating that SHAP and PDP are correctly identifying model behavior. MOSNet (Fig. 3c) has opposite behavior to DNSMOS and, for each of the selected frequency bands except band 3, increases its prediction as the scale factor increases. SCOREQ (Fig. 3e) behaves similarly to, but more aggressively than, DNSMOS. For each of the selected frequency bands, any increase in scale factor resulted in a lower quality prediction, to the extent that by scale factor 1000, SCOREQ outputs only MOS ratings near one. For all three models, a scale factor of less than one had little impact on model predictions.

A central focus of this work is to investigate not only network behavior, but also effects on human perception. Eight randomly-selected evaluation signals (two from each true MOS category) and four scale factors (0.5, 1, 10, 1000) were presented to human subjects in a randomly-ordered pair (original, perturbed). Figures 3b, 3d, and 3f show the effect of perturbation on human ratings. Note that the human responses do not differ between models, but the frequency bands shown are grouped by model for easier comparison. Across all frequency bands, participants rated the perturbed signals as having lower quality than the original. This is likely due to the naivete of perturbation. It is well-known that frequency bands within an audio signal are highly correlated [54] and only perturbing a single band likely results in the addition of perceived noise.

To quantify the agreement between human and model quality predictions, Table 2 gives the Pearson correlation across all scale factors for each frequency band and model. The effects of perturbation on human ratings strongly correlated with that of DNSMOS for frequency bands 3–5, 14, and 31. For SCOREQ, strong correlations were found for all selected frequency bands except band 41. MOSNet exhibited a negative correlation, significant in bands 31 and 41. Thus, DNSMOS and SCOREQ mostly react to modified input in a similar way to humans.

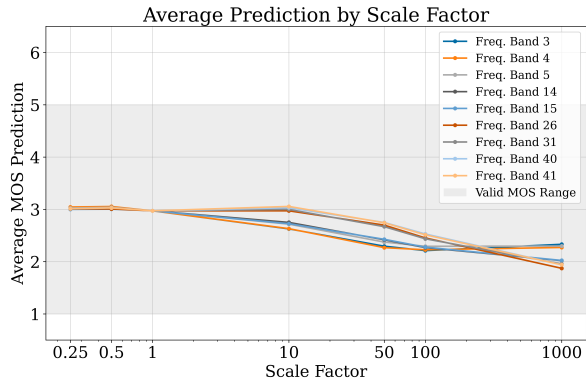
4.4. Qualitative Human Response Data

In addition to providing MOS ratings and signal preference, participants selected whether each of the four provided factors positively, negatively, or did not affect their rating. Figure 4 shows the proportion of positive, negative, and neutral responses for each factor. Compared to the other factors, distortion had the strongest negative impact with 59.8% of participants reporting that it hurt their quality ratings. Conversely, the strongest positive response, with 35.8% of participants, was for SNR, though SNR is also the factor for which responses were most evenly distributed between hurt, helped, or did not affect. For all factors, the positive responses are a strong minority, especially for noise, distortion, and noise type. While this could indicate that humans are more sensitive to detrimental factors, as we saw in Fig. 1, it's also possibly due to how participants interpreted the factor. For example, a positive noise response should indicate that the absence of noise positively affected a participant's rating, however, many participants likely view noise as either neutral or detrimental. Similarly, distortion and noise type may also be seen as negative-only factors. In all cases, positive responses comprise the minority which supports the idea that humans, like networks, are more sensitive to negative factors.

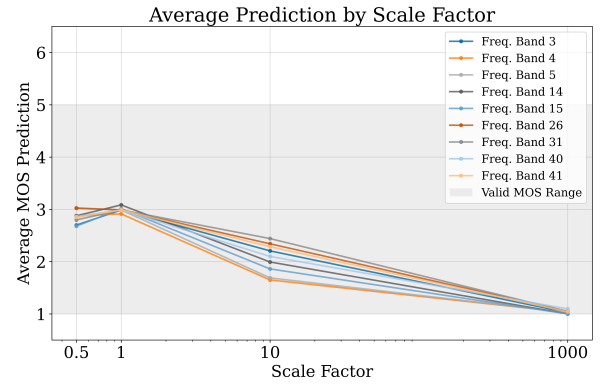
Figure 5 shows the average MOS rating across all signals for which participants selected the (factor, response) pair. For example, the top-left cell indicates that for signals where participants selected that distortion decreased their quality rating, the average MOS was 2.75. The highest average MOS (3.27) occurred when participants reported that noise and distortion positively influenced their ratings, while the lowest average MOS (2.64) occurred when participants reported that SNR negatively influenced their ratings. This figure shows that participants are accurately reporting the factors that affected them since the lowest average MOS rating is given by the negative response and the highest by positive responses for each factor. Additionally, this figure supports the claim that humans are more sensitive to detrimental factors as the difference between average MOS for neutral vs. negative responses is larger than between neutral and positive responses.

5. Discussion

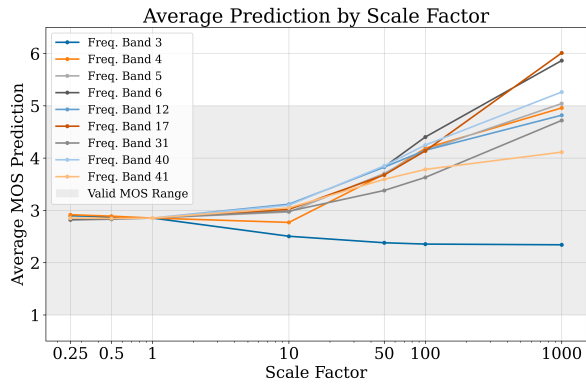
In each of the three experiments MOSNet behaved contrary to DNSMOS and SCOREQ, particularly in the perturbation experiment (Fig. 3c) where MOSNet even predicts invalid MOS ratings that are greater than 5. Though trained on a dataset which has MOS ratings in the range $[1, 5]$, MOSNet does not restrict its prediction, which interestingly contradicts the behavior seen in [9] where they noted that MOSNet predicted only a narrow range of MOS values. As in other works which use MOSNet as a baseline [9, 55, 56], MOSNet had the worst performance of the three models on IUCOSINE (Tab. 1). MOSNet's reliance on low frequencies (Fig. 1b) could explain the model's underperformance in both our and others' works. It's also likely that



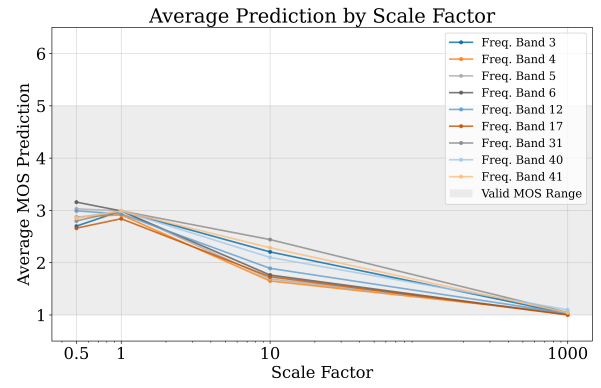
(a) *DNSMOS – Network Behavior*



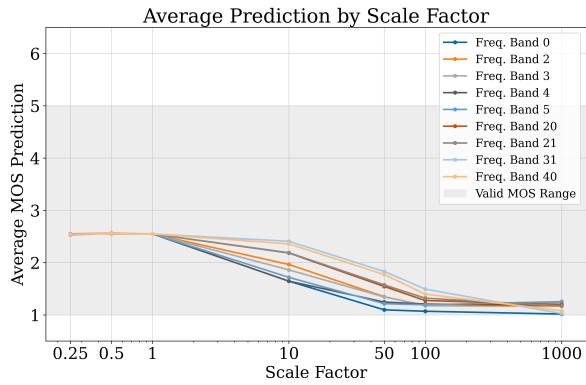
(b) *DNSMOS – Corresponding Human Ratings*



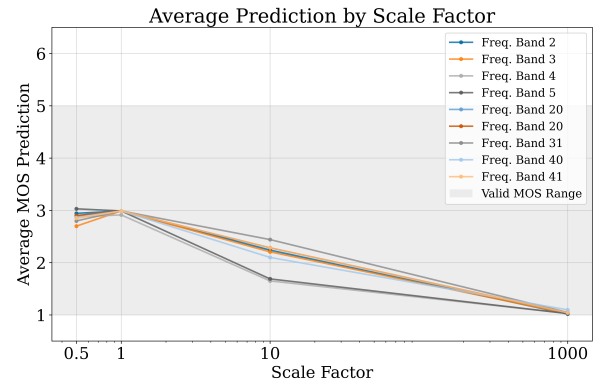
(c) *MOSNet – Network Behavior*



(d) *MOSNet – Corresponding Human Ratings*



(e) *SCOREQ – Network Behavior*



(f) *SCOREQ – Corresponding Human Ratings*

Figure 3: *Effect of perturbation on DNSMOS, MOSNet, and SCOREQ network behavior (left; 3a, 3c, 3e) and human ratings (right; 3b, 3d, 3f). Averages are computed over each of the 100 evaluation files.*

Table 2: Pearson correlation between each model’s perturbation data and human perturbation ratings for each frequency band of interest. Correlations are only given for the nine frequency bands of interest for each model, as discussed in subsection 2.4. Correlations that are statistically significant ($p < 0.05$) are in bold.

Model	Correlation	Frequency Band														
		2	3	4	5	6	12	14	15	17	20	21	26	31	40	41
DNSMOS	PCC		.951	.976	.964			.965	.949				.946	.961	.876	.918
	p-value		.049	.023	.036			.035	.051				.055	.039	.124	.082
MOSNet	PCC		.927	-.744	-.812	-.845	-.903			-.835				-.980	-.933	-.980
	p-value		.073	.256	.188	.155	.097			.165				.020	.067	.020
SCOREQ	PCC		.998	.974	.998	.999					.996	.997		.984	.942	.974
	p-value		.002	.001	.002	.001					.004	.003		.016	.058	.026

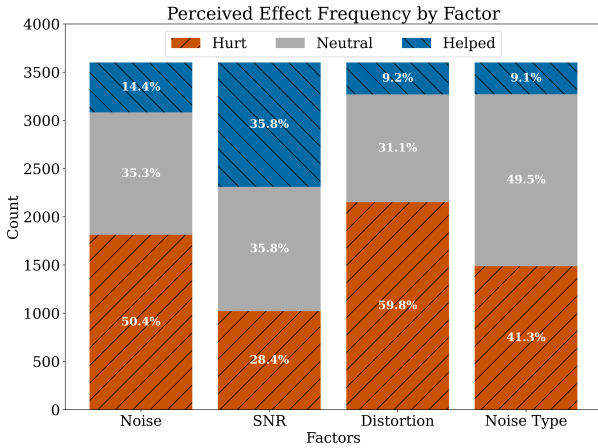


Figure 4: Frequency of response type by factor.

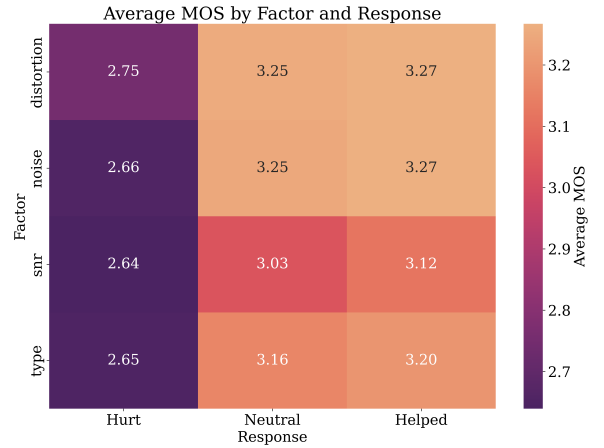


Figure 5: Average MOS by (factor, response type) pair.

this behavior is affected by the architecture we chose to use. We evaluated the CNN-only variant of MOSNet which, unlike the other, does not include bidirectional long short-term memory (BLSTM). We originally chose to use the CNN-only variant of MOSNet to facilitate integration with other explanation methods not explored in this work. Future work using the CNN-BLSTM variant of MOSNet could shed light on MOSNet’s differences from the other networks and provide insight into the functionality that the BLSTM module provides.

All presented explanations describe network behavior as a function of its input, namely frequency bands. There are several explanation methods that focus instead on aspects of the network’s architecture (e.g., activation functions [28], layer gradients [23], etc.). Incorporating other (white-box) explanation approaches that describe behavior based on the network’s inner workings as opposed to a single input feature would provide a fuller picture of network behavior and remains as future work.

Our work is a vital first step towards a unified framework for explaining neural networks in conjunction with human perception. For AI developers, considering the explainability of multiple networks to understand general trends, plus the trends in human ratings that networks should be mimicking, allows them to be intentional about how they develop future networks and benchmarks. For example, we found that a possible cause for MOSNet’s less-than-optimal performance could be because it focuses on a smaller portion of the input than DNSMOS and SCOREQ. This motivates changes to MOSNet, such as inclusion of the BLSTM module, that would allow MOSNet to consider more of its input. For speech and hearing scientists, under-

standing what factors humans focus on and how those factors affect model behavior can motivate dataset generation. Ultimately, we hope this work emphasizes the importance of considering both network and human behavior from several angles when considering the future of explainability.

6. Conclusion

In this work, we evaluated three neural networks, DNSMOS, MOSNet, and SCOREQ for speech quality assessment using perturbation-based explanation techniques and conducted a listening study to determine if humans exhibit the same behaviors. All networks relied on a small portion of the input corresponding to low frequency bands, but MOSNet was particularly focused on a narrow set of bands. Additionally, we found that MOSNet often predicts higher MOS ratings when a given input band has a higher average magnitude, in contrast to DNSMOS and SCOREQ. All three networks are more sensitive to input features which lower the resulting prediction and the same pattern was found among human raters.

Understanding what led to a model’s prediction is not only vital for building trust between humans and neural networks, but important for future development of AI. When a network has lower prediction accuracy than the developers would like, their first step should be asking *why* the accuracy is low and what motivated the network towards its decision. Since the goal of automated speech quality assessment is to match human perception, it is critical that the conversation around model explainability considers how humans determine quality scores.

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8. Generative AI Use Disclosure

No generative AI tools were used in the preparation of this manuscript.

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